



Report 1

Breast-cancer-prediction

Dataset = WISCONSIN DIAGNOSIS BREAST CANCER(WDBC)

Friday, 14th june 2024

Content:

Steps:

1. no outlier removing.
 2. no scaling.
 3. splitting data into train, validation and test using two methods(Random Split, Stratified Split)
 4. Model evaluation(test, validation)
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Data Loading

```
# download from UCI repository

data=
pd.read_csv('https://raw.githubusercontent.com/Muhd-Shahid/Breast-Cancer-Wisconsin/master/data_breast-cancer-wiscons.csv')

data.shape
```

(569, 33)

```
#Removing 'id', 'Unnamed: 32'

data.drop(["id", "Unnamed: 32"], axis = 1, inplace = True)

# Label Encoding

LabelEncoder_dia=LabelEncoder()

data['diagnosis'] = LabelEncoder_dia.fit_transform(data['diagnosis'])
```

Train, Validation, Test

```
random_state = 18

X=data.drop(columns = 'diagnosis')

y=data['diagnosis']
```

Random Split:

```
#Random Split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size =
0.3, random_state = 18)
X_train, X_val, y_train, y_val = train_test_split(X_train, y_train,
test_size = 0.5, random_state = 18)
```

<i>Algorithms</i>	<i>LOG</i>	<i>SVM</i>	<i>RF</i>	<i>KNN</i>	<i>NB</i>	<i>DT</i>
<i>train-test-validation split</i>						
70-15-15 Valid/Test	92.46% ----- 92.98%	89.4% ----- 84.79%	95.4% ----- 94.15%	92.96% ----- 88.88%	95.4% ----- 92.98%	92.96% ----- 90.06%
80-10-10 Valid/Test	92.98% ----- 92.98%	91.66% ----- 87.72%	95.61% ----- 94.73%	93.42% ----- 92.10%	92.54% ----- 92.98%	92.98% ----- 90.35%
90-5-5 Valid/Test	95.31% ----- 91.23%	93.75% ----- 91.23%	92.97% ----- 96.49%	95.31% ----- 92.98%	95.31% ----- 92.98%	90.62% ----- 91.23%
60-20-20 Valid/Test	95.32% ----- 91.66%	88.30% ----- 87.72%	97.07% ----- 96.49%	93.57% ----- 91.66%	95.32% ----- 94.74%	93.57% ----- 92.10%
75-15-10 Valid/Test	96.49% ----- 94.40%	91.22% ----- 86.71%	94.15% ----- 95.10%	92.40% ----- 90.21%	93.57% ----- 94.40%	92.40% ----- 90.90%

Stratified Split:

```
#Stratified Split:
X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.3, stratify=y, random_state=18)

X_train, X_val, y_train, y_val = train_test_split(X_train, y_train,
test_size=0.5, stratify=y_train, random_state=18)
```

<i>Algorithms</i> <i>train-test-validation split</i>	<i>LOG</i>	<i>SVM</i>	<i>RF</i>	<i>KNN</i>	<i>NB</i>	<i>DT</i>
70-15-15 Valid/Test	94.97% ----- 95.32%	92.46% ----- 92.98%	93.97% ----- 93.56%	93.97% ----- 88.88%	93.46% ----- 94.73%	92.96% ----- 91.81%
80-10-10 Valid/Test	93.42% ----- 93.86%	92.54% ----- 92.98%	95.61% ----- 98.24%	92.98% ----- 89.47%	94.73% ----- 97.37%	92.54% ----- 90.35%
90-5-5 Valid/Test	96.09% ----- 98.24%	89.45% ----- 92.98%	96.87% ----- 100.0%	93.75% ----- 91.23%	94.14% ----- 100.0%	91.01% ----- 98.24%
60-20-20 Valid/Test	94.15% ----- 94.30%	91.23% ----- 88.60%	94.74% ----- 96.05%	94.15% ----- 91.66%	92.40% ----- 95.17%	90.64% ----- 92.10%
75-15-10 Valid/Test	91.23% ----- 95.80%	90.06% ----- 91.60%	91.81% ----- 97.20%	91.23% ----- 93.01%	90.06% ----- 96.50%	90.06% ----- 95.80%

Summary:

- The **stratified splitting approach** is generally **preferred over** the **random splitting approach**, as it provides more consistent and representative estimates of model performance, especially for classification tasks where the class balance is important.
- The **90-5-5 stratified split** appears to have the best overall performance and the **most representative** validation and test sets. The **75-15-10 stratified split** is also a strong contender, with consistent and high-performing results. The other split ratios (70-15-15 and 80-10-10) show slightly lower and more variable performance, indicating that they may not be as effective in maintaining the data distribution in the validation and test sets.